Steven Hoekstra

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Profile

I study how household balance sheets (income, liquid and illiquid wealth) shape consumption and macroeconomic forecasts, and when estimated macroeconomic models can be trusted. I embed optimising two-asset households in agent-based models calibrated to administrative microdata, and solve and estimate them with neural networks and simulation-based inference.

Research Interests

Household balance sheets and consumption, heterogeneous-agent models, agent-based modelling, forecasting, expectations, model misspecification, simulation-based inference, housing.

Working Papers and Research Software

Forecasting with a Dutch Agent-Based Model and Optimally Behaving Households
Working paper, with Cars Hommes and Frank Pijpers. PDF
Runner-up, Society for Computational Economics Graduate Student Paper Award, CEF 2026

Identification, Estimation and Misspecification and Their Effects on Forecasting and Policy: Wage–Price Pass-Through in a Dutch Agent-Based Model
Working paper; single-author. PDF. Neural simulation-based inference; known-truth simulation study and application to the 2021–23 Dutch inflation episode, with consequences for forecasts and policy.

BeforeIT.jl: High-Performance Agent-Based Macroeconomics Made Easy
Research software; package co-author. Developed the validation suite, the CANVAS implementation and the multicountry forecasting pipeline.
Package · Original preprint (2025)
Also contributor to CalibrateBeforeIT.jl, calibration routines for BeforeIT.jl models.

Work in Progress

Shortage or Misallocation? The Welfare Loss from Dutch Housing Policy, 2015–2035
Single-author. Long-run welfare loss from housing shortage and mismatch under current Dutch policy, relative to first- and second-best benchmarks, across four levers: the mortgage interest deduction, rent regulation, nitrogen limits on construction and grid congestion. Spatial macroeconomic ABM with housing, labour mobility, demography and firm dynamics; calibration and welfare comparisons are ongoing.

When Does Microdata Help? Conditioning Information in Estimated Structural Models
Single-author. Early stage.

Work Experience

De Nederlandsche Bank — PhD Intern Apr. — Dec. 2025
- Implemented the Bank of Canada CANVAS agent-based model for the Netherlands
- Estimated inflation channels using neural ratio estimation
- Wrote a note on the wage–price channel using this estimated model

Statistics Netherlands (CBS) — Thesis Intern Mar. — Sept. 2023
- Calibrated the agent-based model of Poledna et al. (2023) with Dutch register and national accounts data
- Developed an extension incorporating optimal consumption
- Ran simulations with both models and analysed their internal dynamics

SEO Amsterdam Economics — Research Assistant 2020 — 2021
- Studied academic pressure in secondary schools through stakeholder interviews and a nationwide survey

Education

PhD in Economics — University of Amsterdam 2023 — current
Supervisors: Prof. dr. C.H. Hommes and Prof. dr. F.P. Pijpers
Expected thesis submission: May 2027; expected defence: September 2027

MPhil Economics and Econometrics — Tinbergen Institute ***(cum laude)*** 2021 — 2023
Specialisation: Complexity, Mathematics, and Computation
Thesis: An Agent Based Model of the Dutch Economy
Supervisors: Prof. dr. C.H. Hommes and Prof. dr. F.P. Pijpers

MSc Econometrics — University of Amsterdam ***(cum laude)*** 2020 — 2021
Specialisation: Complexity and Economic Behaviour
Thesis: Endogenous Financial Instability via Heterogeneous Beliefs
Supervisor: Prof. dr. C.H. Hommes

MSc Economics — London School of Economics 2019 — 2020
Specialisation: Public Economics
Thesis: The Inequality of Education using a Two Agent New Keynesian Model
Supervisor: Prof. dr. F. Cowell

Exchange Programme — Royal Melbourne Institute of Technology 2018 — 2019
Minor in Finance and Programming

BSc Economics and Business Economics — University of Groningen 2016 — 2019
Thesis: The Impact of News Coverage on Security Prices

BSc Econometrics — University of Groningen 2015 — 2018
Thesis: Arbitrage Smoothing and Term Structure Models

Grants

NWO Compute Grants — Snellius
Three awarded compute grants of 1,000,000 CPU hours each on Snellius, the Dutch national supercomputer, worth about 15,000 euros each at SURF's rates (3,000,000 CPU hours in total).

Teaching

Graduate

Behavioural Macro and Finance — Lecturer 2025 — present
Developed and taught the MSc lecture series with Cars Hommes, University of Amsterdam.

MPhil guest lecture — Tinbergen Institute
Annual guest lecture in Behavioural Macroeconomics and Complexity.

Undergraduate

BSc Econometrics thesis supervision — University of Amsterdam
Ongoing.

Mathematics 3: Advanced Linear Algebra — Tutor 2025 — 2026

Mathematics 2: Linear Algebra — Tutor 2024 — 2025

Microeconomics for AE — Tutor 2023 — 2024

Presentations

2026	Computing in Economics and Finance (CEF), Society for Computational Economics
2026	Workshop on Economic Science with Heterogeneous Interacting Agents (WEHIA)
2025	Agent-Based Modelling for Policy (ABM4Policy), European Central Bank, Frankfurt
2025	7th Behavioral Macroeconomics Workshop, University of Bamberg
2025	MacroABM 2nd Workshop, *Microdata in Macromodels*, Vienna University of Economics and Business
2025	Workshop on Economic Science with Heterogeneous Interacting Agents (WEHIA)
2025	De Nederlandsche Bank, Amsterdam
2023	De Nederlandsche Bank, Amsterdam

Honours and Scholarships

Honours

Runner-up — Graduate Student Paper Award 2026
Society for Computational Economics, Computing in Economics and Finance (CEF), for Forecasting with a Dutch Agent-Based Model and Optimally Behaving Households.

Cum Laude — MPhil Economics and Econometrics, Tinbergen Institute 2023

Cum Laude — MSc Econometrics, University of Amsterdam 2021

Scholarships

Full scholarship (merit based) — Tinbergen Institute 2022

Tuition fee waiver — Tinbergen Institute 2021 — 2023

Visits and Summer Schools

Visiting Scholar — IIASA, Laxenburg Summer 2022
International Institute for Applied Systems Analysis. Visited Dr. S. Poledna to work on agent-based models

Tools for Macroeconomists: Advanced Tools — University of Oxford Aug. 2021
Doctoral-level course on DSGE models: solving and simulating models with heterogeneous agents, continuous time models, occasionally binding constraints and bounded rationality

Tools for Macroeconomists: The Essentials — University of Oxford Aug. 2021
Doctoral-level course on DSGE models: numerical analysis, perturbation and projection methods, Blanchard-Kahn conditions and sunspot solutions, the Kalman filter and Bayesian estimation

Skills

Programming	Python, Matlab, R, Julia, C, SQL, Bash
High-performance computing	Parallel computing (MPI, BSPlib, Slurm)
Methods	Agent-based modelling, heterogeneous-agent models, neural-network solvers, simulation-based inference, network methods, machine learning, econometrics, dynamic stochastic general equilibrium modelling (Dynare)

References

Prof. dr. C.H. Hommes
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Bank of Canada, Tinbergen Institute
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Prof. dr. F.P. Pijpers
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